Investigation: PETROLEO BRASILEIRO S.A. PETROBRAS (CVM code 9512), fiscal year 2011, as it was on 2024-12-31

Statistical signal based on public financial statements. Not an accusation, not investment advice.

Company

Sector: Petróleo e Gás (origin: FCA); group: non-financial.
Trading codes (2023 FCA, version 1, filed on 2023-03-30): PETR3 (Ações Ordinárias, Bolsa, since 2018-05-14); PETR4 (Ações Preferenciais, Bolsa, since 2018-05-14).

Document used

2011 DFP, version 2, filed on 2012-02-14; origin: open data; type: consolidated; original scale: MIL.
Versions known on the date: 1 (2012-02-09); 2 (2012-02-14).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=14307

Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: −2.458 (index)
- Alert by the study 1 rule: no
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: fewer than 10 companies with a computable value
- Percentile in group and year: 53 (n = 371)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

DSRI, days' sales in receivables index
- Value: 1.072 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: fewer than 10 companies with a computable value
- Percentile in group and year: 60 (n = 467)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

GMI, gross margin index
- Value: 1.138 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: fewer than 10 companies with a computable value
- Percentile in group and year: 79 (n = 477)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

AQI, asset quality index
- Value: 0.895 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: fewer than 10 companies with a computable value
- Percentile in group and year: 19 (n = 472)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

SGI, sales growth index
- Value: 1.153 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: fewer than 10 companies with a computable value

Percentile in group and year: 62 (n = 478)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DEPI, depreciation index
Value: 1.006 (index)
M-score component; it has no rule of its own
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 61 (n = 385)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

SGAI, SG&A expenses index
Value: 0.933 (index)
M-score component; it has no rule of its own
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 31 (n = 471)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

TATA, total accruals to total assets
Value: −0.039 (index)
M-score component; it has no rule of its own
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 52 (n = 494)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

LVGI, leverage index
Value: 1.127 (index)
M-score component; it has no rule of its own
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 70 (n = 486)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DPO jump
Value: 1.9 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 35.1 days)
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 51 (n = 449)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: −6.92 p.p.
Alert by the study 1 rule: no
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 6 (n = 438)
The company's own history: fewer than 5 previous fiscal years with a value

Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: −0.042 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 51 (n = 494)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

Financial risk (context only)

Altman Z'' (financial risk, context only)
Value: 3.050 (index)
Components: X1 = 0.088; X2 = 0.205; X3 = 0.074; X4 = 1.245 (indices)
Percentile in sector and year: 67 (n = 613); percentile in the non-financial universe: the sector has fewer than 10 companies with a value
The company's own history: fewer than 5 previous fiscal years with a value
- No cut-off zone: the available source does not define zones for Z''
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 4/9 (9 computable)
Signals:
- Return on assets (ROA): 1
- Operating cash flow on assets (CFO): 1
- Change in ROA (ΔROA): 0
- Cash above earnings (ACCRUAL): 1
- Change in leverage (ΔLEVER): 0
- Change in current ratio (ΔLIQUID): 0
- No equity issuance (EQ_OFFER): 1
- Change in gross margin (ΔMARGIN): 0
- Change in asset turnover (ΔTURN): 0
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

Changes

Highlighted in the following year's comparative: none.
Highlighted: differences in totals and key lines (total assets, current liabilities, non-current liabilities, equity, net revenue, net income and operating cash flow) above 1% of total assets of the version used. The other differences (reclassifications between lines and smaller differences) stay in the full list.
Full list: 46 accounts above the R$1,000 tolerance (group a, totals and key lines: 2; group b, reclassifications: 44), in the annex at the end.

What was known on the date

- FCA 2011, v1: 2011-02-08
- DFP 2010, v1: 2011-02-25

- DFP 2010, v2: 2011-03-04
- DFP 2010, v3: 2011-03-04
- FCA 2011, v2: 2011-05-09
- FCA 2011, v3: 2011-06-01
- DFP 2011, v1: 2012-02-09
- DFP 2011, v2: 2012-02-14
- DFP 2012, v1: 2013-02-04

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

Data quality

53 applicable checks; 0 failed.

Auditor's report and auditor (factual section; no alert)

Auditor's report: Sem Ressalva (unqualified).
Excerpt checked literally: "Em nossa opinião, as demonstrações contábeis individuais acima referidas apresentam adequadamente, em todos os aspectos relevantes, a posição patrimonial e financeira da Petróleo Brasileiro S.A. - Petrobras em 31 de dezembro de 2011, o desempenho de suas operações e os seus fluxos de caixa para o exercício findo naquela data, de acordo com as práticas contábeis adotadas no Brasil."
Auditor in the 2021 FCA: KPMG Auditores Independentes.
Auditor in the 2022 FCA: KPMG Auditores Independentes.
Auditor in the 2023 FCA: KPMG Auditores Independentes, KPMG Auditores Independentes.
Auditor changes (one line per change, canonical name by the firm's CVM code):

- 2012: KPMG Auditores Independentes -> PricewaterhouseCoopers Auditores Independentes. The outgoing firm audited 6 consecutive fiscal years (2006 to 2011), by the start and end dates in the FCA.
- 2017: PricewaterhouseCoopers Auditores Independentes -> KPMG Auditores Independentes. The outgoing firm audited 5 consecutive fiscal years (2012 to 2016), by the start and end dates in the FCA. The change coincides with the end of the 5-fiscal-year term and is compatible with the rotation rule (art. 31 of CVM Instruction No. 308/1999, in force when the new firm started).

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 39 DFP lines, each with document, version, filing date, account and location.

Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E). Generated at 2026-11-10T19:38:27+00:00.

Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.

Annex: full list of differences in the following year's comparative (R$ million)

(a) BPA 1 Ativo Total: 599,150.0 in the version used, 600,097.0 in the comparative (difference of 947.0)
(a) BPP 2.02 Passivo Não Circulante: 198,714.0 in the version used, 199,661.0 in the comparative (difference of 947.0)
(b) DFC (indirect method) 6.02.08 Investimentos em Títulos e Valores Mobiliários: 11,606.6 in the version used, 0.0 in the comparative (difference of −11,606.6)
(b) DFC (indirect method) 6.02.09 Dividendos Recebidos: 680.5 in the version used, 11,606.6 in the comparative (difference of 10,926.2)
(b) DFC (indirect method) 6.01.01.08 Var. Camb. Monet. e Enc. sobre Financiamentos: 6,237.9 in the version used, 2,892.8 in the comparative (difference of −3,345.1)
(b) DFC (indirect method) 6.01.01 Caixa Gerado nas Operações: 68,071.4 in the version used, 70,964.2 in the comparative (difference of 2,892.8)
(b) DFC (indirect method) 6.01.02 Variações nos Ativos e Passivos: −9,992.4 in the version used, −12,885.2 in the comparative (difference of −2,892.8)
(b) DFC (indirect method) 6.01.02.05 Plano de Pensão e Saúde: 1,483.0 in the version used, −1,409.8 in the comparative (difference of −2,892.8)
(b) BPP 2.01.01 Obrigações Sociais e Trabalhistas: 3,182.1 in the version used, 4,742.2 in the comparative (difference of 1,560.1)
(b) BPP 2.01.01.02 Obrigações Trabalhistas: 2,745.6 in the version used, 4,305.7 in the comparative (difference of 1,560.1)
(b) BPP 2.01.05 Outras Obrigações: 12,250.7 in the version used, 10,690.5 in the comparative (difference of −1,560.1)
(b) BPP 2.01.05.02 Outros: 11,416.4 in the version used, 9,856.2 in the comparative (difference of −1,560.1)
(b) BPP 2.01.05.02.04 Participações de Empregados e Administradores: 1,560.1 in the version used, 0.0 in the comparative (difference of −1,560.1)
(b) BPA 1.02 Ativo Não Circulante: 477,986.3 in the version used, 478,933.3 in the comparative (difference of 947.0)
(b) BPA 1.02.01 Ativo Realizável a Longo Prazo: 41,187.3 in the version used, 42,134.3 in the comparative (difference of 947.0)
(b) BPA 1.02.01.09.05 Depósitos Judiciais: 2,954.9 in the version used, 3,901.9 in the comparative (difference of 947.0)
(b) BPP 2 Passivo Total: 599,150.0 in the version used, 600,097.0 in the comparative (difference of 947.0)
(b) BPA 1.02.01.08 Créditos com Partes Relacionadas: 148.5 in the version used, 1,031.2 in the comparative (difference of 882.7)
(b) BPA 1.02.01.08.04 Créditos com Outras Partes Relacionadas: 0.0 in the version used, 882.7 in the comparative (difference of 882.7)
(b) BPA 1.02.03 Imobilizado: 342,266.9 in the version used, 343,117.3 in the comparative

(difference of 850.4)
(b) BPA 1.02.03.01 Imobilizado em Operação: 183,530.0 in the version used, 184,380.4 in the comparative (difference of 850.4)
(b) BPA 1.02.04 Intangível: 82,284.0 in the version used, 81,433.6 in the comparative (difference of −850.4)
(b) BPA 1.02.04.01 Intangíveis: 81,334.9 in the version used, 80,484.5 in the comparative (difference of −850.4)
(b) BPA 1.02.04.01.02 Direitos e Concessões: 79,653.9 in the version used, 78,803.5 in the comparative (difference of −850.4)
(b) BPA 1.02.01.09.03 Conta Petróleo e Álcool - STN: 831.9 in the version used, 0.0 in the comparative (difference of −831.9)
(b) BPP 2.02.04 Provisões: 26,853.8 in the version used, 27,533.0 in the comparative (difference of 679.2)
(b) BPP 2.02.04.01 Provisões Fiscais Previdenciárias Trabalhistas e Cíveis: 1,361.5 in the version used, 2,040.7 in the comparative (difference of 679.2)
(b) BPP 2.02.04.01.05 Provisões para Outros Processos: 112.5 in the version used, 791.7 in the comparative (difference of 679.2)
(b) BPP 2.02.02 Outras Obrigações: 2,003.4 in the version used, 2,309.9 in the comparative (difference of 306.4)
(b) BPP 2.02.02.02 Outros: 1,816.3 in the version used, 2,122.7 in the comparative (difference of 306.4)
(b) BPP 2.02.02.02.03 Outras Contas e Despesas a Pagar: 1,816.3 in the version used, 2,122.7 in the comparative (difference of 306.4)
(b) BPA 1.02.01.09 Outros Ativos Não Circulantes: 10,926.8 in the version used, 11,041.9 in the comparative (difference of 115.0)
(b) DFC (indirect method) 6.01.01.09 Imposto Renda e Contrib. Soc. Dif. Líq.: 6,156.9 in the version used, 6,237.9 in the comparative (difference of 81.0)
(b) BPA 1.01.03 Contas a Receber: 21,974.7 in the version used, 22,052.6 in the comparative (difference of 77.9)
(b) BPA 1.01.03.01 Clientes: 16,734.0 in the version used, 16,811.9 in the comparative (difference of 77.9)
(b) BPA 1.01.03.01.02 Subsid. Créditos com Pessoas Ligadas: 4,274.3 in the version used, 4,352.2 in the comparative (difference of 77.9)
(b) BPA 1.01.08 Outros Ativos Circulantes: 4,012.3 in the version used, 3,934.4 in the comparative (difference of −77.9)
(b) BPA 1.01.08.03 Outros: 4,012.3 in the version used, 3,934.4 in the comparative (difference of −77.9)
(b) BPA 1.01.08.03.02 Dividendos a Receber: 77.9 in the version used, 0.0 in the comparative (difference of −77.9)
(b) BPA 1.02.01.03 Contas a Receber: 5,122.2 in the version used, 5,071.4 in the comparative (difference of −50.7)
(b) BPA 1.02.01.03.02 Outras Contas a Receber: 5,122.2 in the version used, 5,071.4 in the comparative (difference of −50.7)
(b) BPP 2.02.03 Tributos Diferidos: 33,268.5 in the version used, 33,229.8 in the comparative (difference of −38.7)
(b) BPP 2.02.03.01 Imposto de Renda e Contribuição Social Diferidos: 33,268.5 in the version used, 33,229.8 in the comparative (difference of −38.7)
(b) BPP 2.02.03.01.02 Outros Impostos Diferidos: 38.7 in the version used, 0.0 in the comparative (difference of −38.7)
(b) BPA 1.02.01.09.04 Investimentos em Empesas Privatizáveis: 2.2 in the version used, 0.0 in the comparative (difference of −2.2)
(b) BPA 1.02.01.09.07 Outros Realizáveis a Longo Prazo: 1,245.9 in the version used, 1,248.2 in the comparative (difference of 2.2)